

Bimetric Edges on a Shared Skeleton: Groundwork for an Eisenstein Dark Sector in Regge Calculus

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Abstract

We lay the contextual groundwork for a discrete bimetric framework in which a single simplicial skeleton carries two length assignments, one baryonic and one dark, the latter generated by the Eisenstein Gram matrix. The off-diagonal entry of that matrix is shown to be the direct origin of the cross-term in the proposed dark dispersion relation $E^2 = p^2 c^2 + (mc^2)(pc) + m^2 c^4$, so that the kinematic proposal and the simplicial geometry are two readings of one bilinear form. We then determine where torsion enters. In the continuum a second metric sources disformation rather than torsion, since the difference of two Levi-Civita connections is symmetric in its lower indices; on the lattice, where the deformation is assigned per simplex and jumps across shared faces, the development of a vertex star tears at exactly those faces whose spoke norms disagree between the sectors, and the tear is a Burgers vector, the discrete signature of torsion. We exhibit the mechanism on the Eisenstein vertex star, where the two sectors assign the same combinatorial star curvatures of opposite sign, and where the tear at a face is simply the difference of the two sectors' lengths for the edge crossing it, vanishing on edges whose Eisenstein and Euclidean norms coincide, so that dark domain walls can be torsion-silent or torsion-loud according to the arithmetic of their boundary edges. The identification of these tears with torsion proper, invariance under the development freedom and consistency with the Drummond construction, is stated as the framework's central open verification rather than asserted. The construction throughout is kinematic: no action is given for the deformation parameter ε , and the dispersion relation is a geometric ansatz rather than a derivation from a Lagrangian or a representation of the Poincaré algebra. §7 states these gaps explicitly and names what would be needed to close them.

1 Two sectors, one skeleton

Every dark matter observation is, at bottom, an inference about spacetime geometry. Rotation curves, gravitational lensing, cluster dynamics, and cosmological structure are not direct sightings of a substance; they are reconstructions from the metric by which geodesics are interpreted, and the dark matter hypothesis is one way of accounting for the mismatch between the geometry those observations imply and the geometry the visible matter should produce. If one wishes to close that mismatch while retaining a single underlying arena, then the most economical object to modify is the metric itself, the rule converting displacements into lengths, and the most conservative such modification gives the modified rule only to the sector whose measurements have never been made: the dark one. That is the entire proposal, and it should be said at the outset what this paper is and is not. It is not an alternative cosmology, and it fits no rotation curve. It is a proposal for a geometric language in which dark-sector kinematics, Regge geometry, and discrete torsion become manifestations of one bilinear form, together with an honest ledger of what that language can currently say and what it cannot.

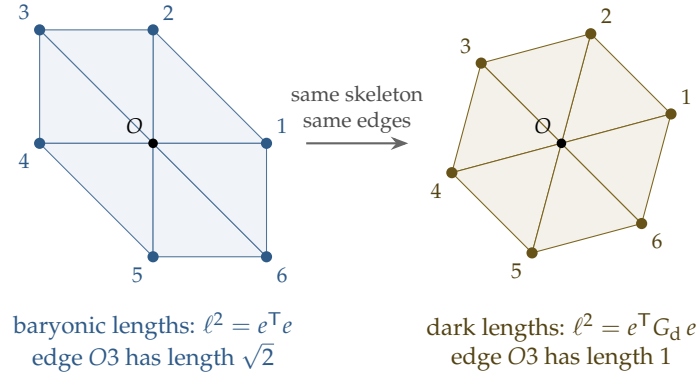


Figure 1: The defining picture of the paper. Left and right show the *same* simplicial complex: identical vertices, identical edges, identical gluing. What differs is the rule each sector uses to convert an edge into a length, the baryonic identity Gram matrix on the left, the Eisenstein Gram matrix on the right (the right panel is drawn in the chart where dark lengths appear as true lengths). Some edges keep their length across the two readings (O1, O2), others do not (O3: $\sqrt{2}$ baryonic, 1 dark), and that selective disagreement is the source of every effect in the paper. There are not two lattices; there is one skeleton and two rulers.

The starting position is deliberately conservative about ontology and radical only about measurement. There is a single spacetime, discretised in the manner of Regge as a simplicial complex $\mathcal{C}$ whose geometry is carried entirely by data on its edges. Baryonic matter and dark matter inhabit the same complex, traverse the same edges, and meet at the same vertices; what distinguishes the two sectors is the rule by which each converts an edge vector into a length (Figure 1). The baryonic sector uses the standard quadratic form, which in the flat interior of any simplex is the Euclidean (or Lorentzian) inner product, while the dark sector uses a second quadratic form whose Gram matrix carries an off-diagonal entry. Everything that follows, the modified dispersion relation, the doubled deficit angles, and the eventual appearance of contorsion-like terms, flows from that single off-diagonal number.

Two ideas carry the whole paper, and neither is assumed familiar. The first is bimetricity, whose conceptual case was made above and whose lineage §2 supplies. The second is Regge calculus, the discretisation of general relativity in which geometry lives entirely on the edges of a simplicial complex; §4 builds it from the deficit angle upward, because the paper's central results, the doubled curvatures and the dislocation torsion, are statements *about* the Regge machinery and cannot be understood without it. Figure 2 shows how the pieces depend on one another; the appendices hold material addressed to the specialist reader, and nothing in the main line depends on them.

Cursory: why a Gram matrix is the whole story

An inner product on $\mathbb{R}^n$ is completely specified by a symmetric matrix G through $\langle u, v \rangle_G = u^T G v$. The matrix entries are the inner products of the basis vectors, $G_{ij} = \langle e_i, e_j \rangle$, so the diagonal records the squared lengths of the basis vectors and the off-diagonal records the angles between them. Changing G changes every length and every angle in the space while leaving the underlying vectors untouched; this is exactly the freedom the dark sector exploits.

The baryonic and dark Gram matrices, written for the moment in two dimensions where the essential structure already lives, are

$$G_b = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad G_d = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \quad (1)$$

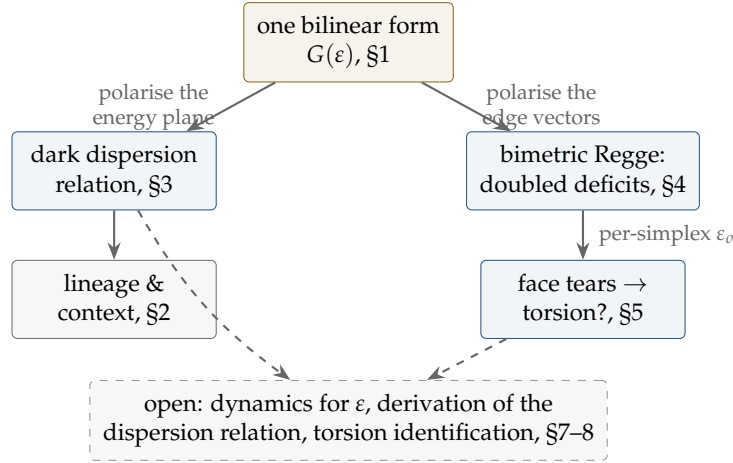


Figure 2: Roadmap. One bilinear form read twice: polarised on the energy plane it yields the dark dispersion relation; polarised on edge vectors it yields the bimetric Regge geometry, whose per-simplex deformation produces the face tears of §5. Solid arrows are derivations within the paper; dashed arrows point at the open problems of §§7–8, on which the framework’s status as physics, rather than geometry, depends.

and it is useful to keep a one-parameter family between them,

$$G(\varepsilon) = \begin{pmatrix} 1 & \varepsilon/2 \\ \varepsilon/2 & 1 \end{pmatrix}, \quad \varepsilon \in [0, 1], \quad (2)$$

so that $\varepsilon = 0$ recovers the baryonic sector and $\varepsilon = 1$ the Eisenstein one. The dark squared length of a vector $v = (x, y)^T$ is then

$$\|v\|_d^2 = v^T G_d v = x^2 + xy + y^2, \quad (3)$$

which is the norm form of the Eisenstein integers $\mathbb{Z}[\omega]$, the ring generated by a primitive cube root of unity. The hexagonal lattice, the A_2 root system of $SU(3)$, and the sixty-degree basis are all encoded in the half.

Remark 1. *It matters for honesty in later sections that G_d is positive definite and hence a legitimate inner product; the Eisenstein form passes the parallelogram law and is therefore, by Jordan–von Neumann, an inner product in different clothes rather than a Finsler norm. A lone inner product on flat space is undetectable, since a linear change of basis maps it to the identity. The physics of this paper lives entirely in the coexistence of two inner products on shared coordinates, which is the defining situation of a bimetric theory.*

2 Lineage: whose two metrics

2.1 From motivation to history

The conceptual case for bimetricity was made in §1: dark-sector evidence is geometric inference, and a sector-restricted second metric is the minimal modification that touches measurement while leaving the arena single and shared. History is not motivation, but it is due diligence, and a proposal that walks a century-old road should know where the road has already gone and where its predecessors fell. This section supplies that, and then locates the present construction’s departures.

2.2 The bimetric lineage

The idea of a second metric is nearly as old as general relativity's maturity. Rosen introduced a flat background metric alongside the dynamical one in 1940 [1], and his later bimetric theory of gravitation [2] survived solar-system tests only to be excluded by binary-pulsar energy loss, an early lesson that a second metric earns its keep or dies by its phenomenology. Isham, Salam and Strathdee's f-g gravity [3] coupled a second tensor field to the hadronic sector specifically, and is the closest historical ancestor of the present proposal's sector-restriction: different matter, different metric. The modern era descends from the massive-graviton problem: the Fierz-Pauli linear theory [4], the Boulware-Deser ghost that haunted its nonlinear completions [5], the de Rham-Gabadadze-Tolley resolution [6, 7], and finally Hassan and Rosen's promotion of the reference metric to a dynamical field, yielding ghost-free bimetric gravity [8], reviewed in [9]. Within that framework the massive spin-2 field itself has been proposed as the dark matter [10], while on the modified-dynamics side both Bekenstein's TeVeS [11] and Milgrom's explicitly bimetric MOND [12] use additional metric structure to do work usually assigned to dark particles. The kinematic face of the proposal, a sector-specific dispersion relation, has its established formalism in the Standard Model Extension [13], which catalogues Lorentz-violating operators including terms bilinear in mass and momentum, and in the quantum-gravity phenomenology of modified dispersion relations [14]; geometrically, the licence to consider non-orthogonal norms at all is the Finsler programme [15], which the present construction under-shoots deliberately, since the Eisenstein form remains an inner product.

2.3 Where the Gram-matrix version stands apart

Against this lineage the present construction makes three departures. First, the second metric is not a dynamical spin-2 field: it is algebraically rigid, fixed by the Eisenstein norm form up to the single interpolation parameter ε , so the ghost problematic that shaped the modern literature is sidestepped rather than solved, there being no second kinetic sector to propagate a ghost. The cost of that rigidity is honesty about what is dynamical, nothing beyond the shared edge lengths in the minimal version, the per-simplex ε_σ in the dynamical extension of the closing programme; the benefit is predictivity, since a one-parameter deformation cannot be tuned region by region to fit rotation curves. Second, the deformation has an arithmetic origin rather than an effective-field-theory one: the coefficient of the dispersion cross-term is an order-one number fixed by lattice geometry rather than a small Wilson coefficient awaiting bounds, which is why the proposal must be confined to the dark sector, where the orthogonal relation is untested, and why it is falsifiable there in a way Planck-suppressed modifications are not. Third, the natural home is discrete: where the bimetric literature works on smooth manifolds, the present framework is built on the Regge lattice [16] from the start, and §5 shows the discreteness is load-bearing, since the torsion content of the theory exists only there. The discrete side has its own lineage this paper joins: Drummond's piecewise-constant torsion fields on the simplicial lattice [17], the Poincaré gauge formulation of Regge calculus by Caselle, D'Adda and Magnea [18], the identification of torsion with dislocations and curvature with disclinations [19, 20, 21], teleparallel Regge calculus [22], the metric-affine framework housing curvature, torsion and nonmetricity together [23], and the geometric trinity that makes their equivalence precise [24]. The companion note distributes that trinity across the two sectors.

3 From the off-diagonal entry to the dark dispersion relation

The proposed dark-sector dispersion relation replaces the orthogonal Pythagorean combination of momentum energy and rest energy with the Eisenstein combination,

$$E^2 = (pc)^2 + (mc^2)(pc) + (mc^2)^2. \quad (4)$$

The claim to be established here is that equation (4) is nothing more, and nothing less, than the statement that the dark sector measures the energy plane with G_d .

Consider the abstract two-dimensional energy plane with coordinates

$$X = pc, \quad Y = mc^2, \quad (5)$$

so that a particle's kinematic state, insofar as the dispersion relation is concerned, is the vector $v = (X, Y)^T$. The baryonic dispersion relation is the statement that total energy is the baryonic length of this vector,

$$E_b^2 = v^T G_b v = X^2 + Y^2 = p^2 c^2 + m^2 c^4, \quad (6)$$

which is the familiar energy Pythagoras: momentum energy and rest energy contribute at right angles, each blind to the other. The dark dispersion relation is the same sentence with the other Gram matrix,

$$E_d^2 = v^T G_d v = X^2 + XY + Y^2 = p^2 c^2 + (mc^2)(pc) + m^2 c^4. \quad (7)$$

The cross-term is the polarisation of the off-diagonal entry

Expanding the quadratic form of $G(\varepsilon)$ gives

$$v^T G(\varepsilon) v = X^2 + \varepsilon XY + Y^2,$$

so the coefficient of the cross-term is exactly twice the off-diagonal Gram entry. The cross-term of the dark dispersion relation and the sixty-degree angle of the Eisenstein basis are one number seen twice. There is no independent kinematic postulate; the dispersion relation is inherited from the geometry.

Cursory: polarisation, in one line

A quadratic form $Q(v) = v^T G v$ determines the full bilinear form by the polarisation identity $\langle u, v \rangle_G = \frac{1}{2}(Q(u+v) - Q(u) - Q(v))$. This identity is what fails for genuine Finsler norms and what succeeds here; it is also, as §4 shows, the identity on which all of Regge calculus secretly runs, since the law of cosines is polarisation read as trigonometry.

Two immediate checks confirm that the proposal degrades gracefully. At rest, $X = 0$ and $E_d = mc^2$; for a massless particle, $Y = 0$ and $E_d = pc$. The cross-term switches on only when both mass and momentum are nonzero, which is to say only for massive dark particles in motion, and this is precisely the kinematic regime in which dark matter is astronomically inferred and terrestrially unconstrained.

4 Regge calculus and where the Gram matrix enters it

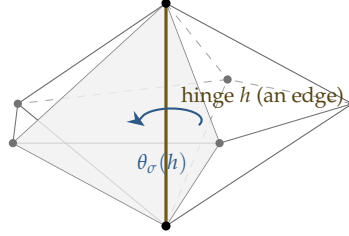
This section is the load-bearing wall of the paper, and a reader meeting Regge calculus for the first time should slow down here, because everything after it, the doubled curvatures, the dislocations, the contorsion, is a statement about the machinery this section builds. The compensation for the effort is that the machinery is small: Regge calculus needs no coordinates, no Christoffel symbols, and no tensor calculus, only lengths, angles, and the observation that curvature can be made to live at isolated places rather than everywhere.

Regge's discretisation of general relativity replaces the smooth manifold by a simplicial complex and the metric tensor by the assignment of a length to every edge. Each simplex is internally flat; curvature cannot hide inside a simplex and is instead concentrated on the hinges, the codimension-two faces (triangles in four dimensions, edges in three, vertices in

two), where the surrounding simplices fail to fill the full angle (Figure 3). The failure is the deficit angle

$$\epsilon_h = 2\pi - \sum_{\sigma \supset h} \theta_\sigma(h), \quad (8)$$

where $\theta_\sigma(h)$ is the dihedral angle of simplex σ at hinge h , and the Regge action is the hinge-weighted sum $S = \sum_h A_h \epsilon_h$, with A_h the hinge volume.



five tetrahedra sharing one edge; each contributes a dihedral angle at the hinge, and curvature is the failure of those angles to sum to 2π

Figure 3: The hinge in three dimensions. In 3D Regge calculus a hinge is an *edge* (gold), and the simplices meeting around it are tetrahedra, five shown here fanned about the shared edge AB . Each tetrahedron is internally flat and contributes one dihedral angle $\theta_\sigma(h)$ measured around the hinge; the deficit $\epsilon_h = 2\pi - \sum_\sigma \theta_\sigma(h)$ is where all the curvature lives. The two-dimensional pictures elsewhere in this paper are the slice of this figure perpendicular to the gold edge, with the hinge appearing as a single vertex.

The decisive structural fact is that every quantity in this action, the dihedral angles and the hinge volumes alike, is computed from edge lengths through the law of cosines, and the law of cosines is polarisation: recovering the angle between two edges from three lengths presupposes that lengths come from a bilinear form. This is the precise doorway through which a second Gram matrix walks into the discrete theory. The operation used repeatedly below, *developing* the complex, unrolling the simplices around a hinge into a single flat chart, is shown as a sequence in Figure 4; it is the most intuitive picture in Regge calculus and the one on which every computation in §5 runs.

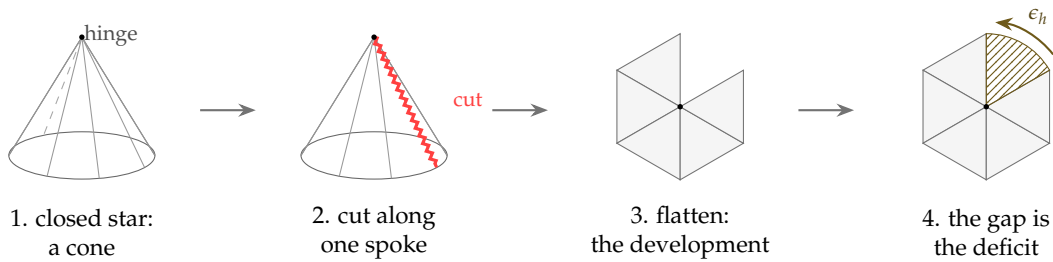


Figure 4: Development, the operation behind every computation in this paper. A closed vertex star is intrinsically a cone (1); cut it along one spoke (2) and unroll the triangles, each internally flat, into a single plane (3); if the star was genuinely curved the unrolled fan fails to close, and the gap (4) is the deficit angle, the curvature the cone was carrying at its hinge. Everything in §5 is this operation performed with two different rulers: when the triangles carry unequal deformations, the fan can fail to close in length as well as in angle, and the length failure is the face tear.

Cursory: deficit angle as curvature

Cut a paper disc, remove a wedge of angle ϵ , and tape the cut edges together; the result is a cone, flat everywhere except at the apex, where all the curvature of the removed wedge now lives. Parallel transport a vector around the apex and it returns rotated by exactly ϵ . Regge curvature is this construction industrialised: every hinge is a potential cone apex, and the deficit angle is the holonomy of a loop around it.

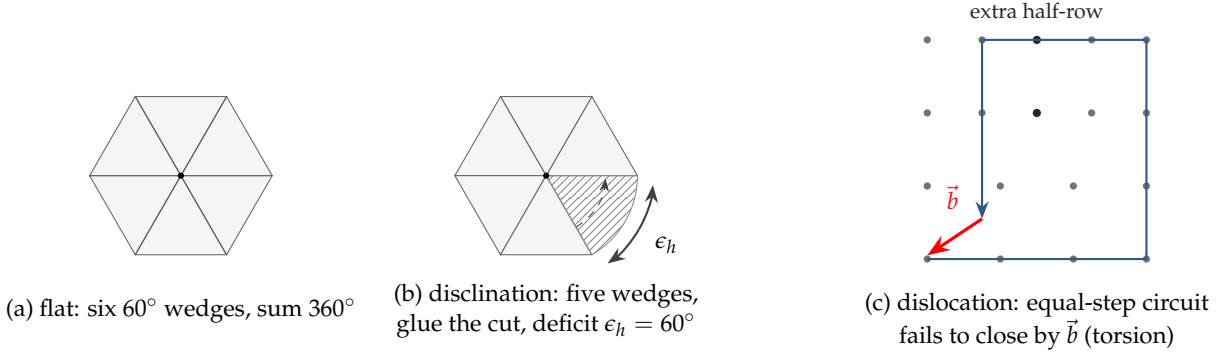


Figure 5: Curvature and torsion as lattice defects in standard Regge calculus. (a) Six 60° triangles close the vertex star exactly: flat. (b) Remove one and glue the cut edges (dashed identification): the star becomes a cone, all curvature concentrated at the hinge as the deficit angle ϵ_h , a disclination. (c) The other defect: around a dislocation, a circuit of equal lattice steps fails to close, and the failure $\vec{b}$ is the Burgers vector, the discrete signature of torsion; in three dimensions $\vec{b}$ may point along the hinge itself, the screw dislocation, which is where contorsion proper lives.

4.1 The bimetric assignment

Give every edge e of the complex a vector $\vec{e}$ in the frame of each simplex containing it, and assign it two squared lengths,

$$\ell_b^2(e) = \vec{e}^\top G_b \vec{e}, \quad \ell_d^2(e) = \vec{e}^\top G_d \vec{e} = \ell_b^2(e) + \varepsilon(e_x e_y), \quad (9)$$

where the final expression, written in two dimensions for clarity, displays the dark deformation as a cross-term between edge components, the simplicial shadow of the kinematic cross-term of equation (4). Each length assignment feeds the Regge machinery independently: baryonic dihedral angles, baryonic deficits, and a baryonic action from ℓ_b ; dark dihedral angles, dark deficits, and a dark action from ℓ_d . The skeleton is shared, so there is one arena; the curvatures are doubled, so the two sectors free-fall differently through it, and a baryonic observer attributing all geodesic motion to the baryonic deficits will infer unaccounted gravitation wherever the dark deficits diverge from the baryonic ones. That inference is the phenomenological content of the proposal.

Remark 2. If edge vectors are restricted to the Eisenstein lattice itself, the dark squared lengths take values in the Loeschian numbers $a^2 + ab + b^2 \in \{1, 3, 4, 7, 9, 12, 13, \dots\}$, giving a principled, arithmetic version of the discrete Regge model, in which edge lengths are restricted to a finite menu for computability. Here the menu is dictated by the ideal theory of $\mathbb{Z}[\omega]$ rather than by numerical convenience.

5 Contorsion in the bimetric lattice

The terms need fixing before the mechanism can be stated, and the statement then splits cleanly by regime: the continuum forbids torsion here, and the lattice supplies it.

5.1 Torsion, contorsion, and the decomposition of a connection

A connection Γ tells vectors how to stay parallel as they move. The torsion of a connection is the antisymmetric part of its lower indices,

$$T^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\mu\nu} - \Gamma^\lambda{}_{\nu\mu}, \quad (10)$$

and its geometric meaning is the failure of infinitesimal parallelograms to close: transport $\vec{u}$ along $\vec{v}$ and $\vec{v}$ along $\vec{u}$, and torsion measures the gap between the two endpoints. In the metric-affine decomposition, a general connection splits into three tensorially independent pieces,

$$\Gamma^\lambda{}_{\mu\nu} = \underbrace{\{\lambda{}_{\mu\nu}\}}_{\text{Levi-Civita}} + \underbrace{K^\lambda{}_{\mu\nu}}_{\text{contorsion}} + \underbrace{L^\lambda{}_{\mu\nu}}_{\text{disformation}}, \quad (11)$$

where the contorsion K is built algebraically from the torsion T and the disformation L is built from the nonmetricity $Q_{\lambda\mu\nu} = \nabla_\lambda g_{\mu\nu}$, the failure of the metric to be covariantly constant.

Cursory: contorsion, plainly

Contorsion is the correction you add to the torsion-free Levi-Civita connection to obtain a connection that twists. Where the Levi-Civita connection transports frames without rotation about the direction of motion, a connection with contorsion screws them as they travel, in the way a barber-pole stripe screws around its cylinder. Einstein–Cartan theory sources this twist from intrinsic spin; the question here is whether a second metric can source it instead.

5.2 The continuum verdict: disformation, not torsion

Suppose the dark sector carries its own metric $g_d = G(\varepsilon)$ smoothly over the manifold and uses its own Levi-Civita connection Γ_d . The difference of two connections is always a tensor,

$$C^\lambda{}_{\mu\nu} = \Gamma_d^\lambda{}_{\mu\nu} - \Gamma_b^\lambda{}_{\mu\nu}, \quad (12)$$

but both connections, being Levi-Civita, are symmetric in $\mu\nu$, and therefore so is C . Its antisymmetric part vanishes identically, and with it any torsion and any contorsion.

Proposition 1. *A bimetric structure in which each sector uses the Levi-Civita connection of its own metric generates no torsion in either sector. Measured against the baryonic connection, the dark metric exhibits nonmetricity, $\nabla^b g_d \neq 0$ wherever ε varies, so the difference tensor C is pure disformation in the decomposition (11).*

This is the honest continuum statement, and it comes with an immediate corollary worth absorbing: if ε is constant, the two Levi-Civita connections coincide on flat space, C vanishes, and the bimetric structure is globally invisible, exactly as the earlier remark about a lone inner product predicted. The deformation must vary for anything to happen. A position-dependent $\varepsilon(x)$ sources disformation, which places the natural continuum home of this proposal in symmetric teleparallel and general metric-affine territory rather than in Einstein–Cartan theory.

5.3 The lattice verdict: dislocations at norm-sensitive faces

The simplicial discretisation reverses the verdict, and the mechanism is constructive. On the Regge lattice the natural bimetric assignment is not a smooth $\varepsilon(x)$ but a piecewise-constant one: each simplex σ carries its own deformation ε_σ , constant on the flat interior, jumping across the shared face when one passes to a neighbour. Developing the complex, unrolling the simplices around a hinge into a single flat chart, requires gluing each simplex to the next along the shared face, and the gluing can always match the face's direction by a rotation about the hinge, while it can match the face's length only when the two simplices' norms agree on it. Where they disagree, the two images of the far vertex split, and the gap

$$b_k = (\text{image of vertex } k \text{ from simplex } \sigma_k) - (\text{image from simplex } \sigma_{k-1}) \quad (13)$$

is a Burgers vector, the same object that measures a dislocation in a crystal and the standard discrete signature of torsion; this is the construction by which Drummond [17], and Caselle, D'Adda and Magnea [18], carried torsion onto the Regge lattice as piecewise-constant fields with face discontinuities. In the present framework the discontinuities are supplied by the physics, since ε_σ jumping across a face is exactly such a field, and the contorsion of the dark sector is the part of the face transition maps generating these closure failures.

Torsion is arithmetically selective

The Burgers vector at a face is the difference of the two sectors' lengths for the spoke crossing it, so it vanishes identically on spokes whose Eisenstein and Euclidean norms coincide, and is nonzero precisely on the norm-sensitive spokes. Among the six Eisenstein units, the four spokes $\pm(1,0)$, $\pm(0,1)$ have squared length 1 in both sectors, while $\pm(-1,1)$ have squared length 2 baryonic and 1 dark. A dark domain wall therefore carries torsion or does not according to the number theory of its boundary edges: the lattice admits torsion-silent walls, invisible in the dislocation content of the complex, alongside torsion-loud ones. Discreteness is the mechanism, and arithmetic is the selection rule.

5.4 A worked example, and the claim that actually needs proving

The mechanism is best seen on the smallest complex that exhibits it, the six-triangle vertex star with per-triangle ε_σ , and it is worth being candid about the status of what follows: these are illustrations of the construction, not findings. The central fact is one line. The tear at a face is, by construction, the difference between the two triangles' lengths for the spoke crossing it,

$$|b| = |\ell_{\varepsilon_\sigma}(e) - \ell_{\varepsilon_{\sigma'}}(e)|, \quad (14)$$

so every property of the tear, its magnitude, its scaling, and its vanishing, is elementary once the norms are written down. For the spoke $(-1,1)$ between a dark and a baryonic triangle this gives $\sqrt{2} - 1$; the small-jump behaviour is the first-order expansion of $\sqrt{2 - \varepsilon}$; and the tear vanishes identically on spokes with $e_x e_y = 0$, where the cross-term of the deformation has nothing to act on, which is all the "arithmetic selectivity" of the previous subsection amounts to at the level of a single face: the deformation is a pure cross-term, so its effects live exactly where edge components mix. None of this is deep, and the worked example earns its place only as a concreteness check that the machinery does what its definitions say.

The claim that carries weight, and that equation (14) does not by itself establish, is the identification of these tears with torsion in the sense of the discrete gravity literature. Two things must hold for that identification to be more than a borrowed word. First, the tear must be invariant under the freedoms of the construction: the choice of development chart, the ordering of the gluing, and the frame identifications across faces, since a quantity that can be gauged

away by re-developing the same complex is bookkeeping, not geometry. The magnitude (14) is manifestly independent of chart and ordering, being a difference of two lengths each computed within a single simplex, but the full Burgers *vector*, direction included, transforms under the development freedom, and the invariant content is properly the conjugacy class of the affine holonomy around the hinge, which has not been computed here. Second, the identification must reproduce the established constructions in their domain: Drummond's torsion [17] and the dislocation picture of Schmidt and Kohler [19] carry specific transformation properties and Bianchi-type constraints, and showing that bimetric tears satisfy them, rather than merely resembling them, is the substantive open task. It stands first in the programme below, ahead of any extension to higher dimensions, because if the identification fails, §5 describes a book-keeping artefact with a suggestive name, and if it succeeds, the framework inherits the full apparatus of discrete Einstein–Cartan theory at the price of one Gram matrix.

The crossing values, for completeness. The deficit of an n -star of fundamental triangles vanishes at $\varepsilon_n = 2 \cos(2\pi/n)$, each an algebraic integer of the real cyclotomic field $\mathbb{Q}(\zeta_n)^+$: $\varepsilon_4 = 0$, $\varepsilon_5 = 1/\varphi$, $\varepsilon_6 = 1$, $\varepsilon_8 = \sqrt{2}$. The endpoints ε_4 and ε_6 being exactly the two sectors restates the design fact that the square and hexagonal closures are the two crystallographic ones; the intermediate values are forced trigonometry of the chosen n and carry no significance.

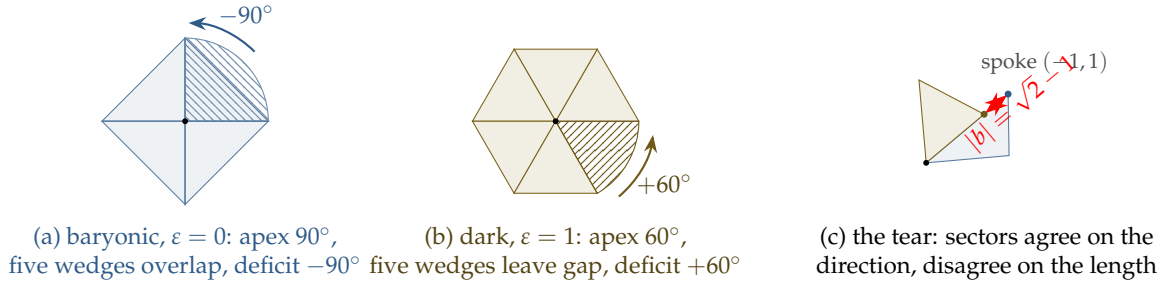


Figure 6: The bimetric view, in the two sector colours. Panels (a) and (b) develop the *same* combinatorial five-triangle star: the baryonic sector (blue) measures each apex at $\arccos(0) = 90^\circ$, so the wedges overlap and the vertex carries deficit -90° ; the dark sector (gold) measures the identical star at $\arccos(1/2) = 60^\circ$, leaving a gap of $+60^\circ$. One skeleton, two curvatures of opposite sign; the crossing value for this star, $\varepsilon_5 = 2 \cos 72^\circ$, is pentagon trigonometry forced by the choice $n = 5$. Panel (c) shows the torsion mechanism: two triangles of unequal ε_σ glued along the norm-sensitive spoke $(-1, 1)$ agree on its direction, which a rotation can always arrange, and disagree on its length, $\sqrt{2}$ against 1, so the far vertex splits by the Burgers vector $|b| = \sqrt{2} - 1$ of equation (13). On norm-agnostic spokes the two colours would meet at the same point and the face would carry no torsion.

5.5 What this entails physically

Three entailments follow. First, the dark sector carries an Einstein–Cartan-like structure without spin sources: contorsion is generated geometrically by the granularity of the deformation field, so torsion marks where the dark deformation is inhomogeneous, concentrated on the boundaries between regions of different ε , and further filtered by the arithmetic selection rule, so that the dislocation content of a region reads off both the geography of the dark domains and the residue classes of their walls. Second, since contorsion couples to fermion spin in Einstein–Cartan theory, any portal interaction between the sectors acquires a natural channel: baryonic fermions traversing a torsion-loud wall feel an effective spin coupling, while torsion-silent walls of identical ε -profile do not produce it, which is a discriminating signature rather than a generic one. Third, the structure is finitely computable throughout: transition maps are

small matrices, dislocations are vector differences, and the full torsion content of a complex is linear algebra over $G(\varepsilon)$.

6 The computational programme

The worked example of §5.4 illustrates the mechanism; the programme ahead must first establish it and then scale it.

1. **The torsion identification.** Compute the affine holonomy around a hinge with mixed ε_σ , extract its conjugacy class, and verify that the translational part transforms as Drummond's discrete torsion and satisfies the corresponding Bianchi-type constraints; this is the load-bearing check named in §5.4, and everything downstream is conditional on it.
2. **Three and four dimensions.** Repeat the development around a hinge (an edge in 3D, a triangle in 4D) with per-simplex ε_σ ; the Burgers vector acquires components parallel to the hinge (screw dislocations) alongside the perpendicular tears (edge dislocations) seen in 2D, and the screw component is where genuinely three-dimensional contorsion lives.
3. **Contorsion extraction.** Decompose the face transition maps in the discrete analogue of equation (11), isolating the antisymmetric generator, and map its concentration across a complex with a prescribed ε -landscape.
4. **Crossing values and lattice compatibility.** The zero-deficit deformations $\varepsilon_n = 2 \cos(2\pi/n)$ single out the crystallographic cases $n = 4, 6$ as the endpoints of the family; the substantive question is not the numerology of the intermediate values but whether any structure of the framework, the Loeschian edge-length menu, the norm-sensitivity selection rule, degenerates or changes character at non-crystallographic ε , where $G(\varepsilon)$ is the Gram matrix of no lattice at all.
5. **Regge dynamics.** Promote ε_σ to a dynamical variable alongside the edge lengths, vary the doubled Regge action, and determine whether domain walls form spontaneously and whether the torsion-silent walls are energetically preferred, which would make the dark sector's dislocation content sparse in a predictable pattern.

7 Limitations: what remains kinematic

The construction assembled in this paper is almost entirely kinematic, and it is worth saying so plainly rather than letting the geometry's fluency obscure it. Every result above concerns metrics, norms, lengths, deficit angles, and Burgers vectors: static structures computed once a value of ε is given. Nothing in the paper says what fixes that value, why it should vary across the complex, or what would happen if it did. Two specific gaps follow from this, and a third concerns the dispersion relation itself.

7.1 No dynamics for ε

The deformation parameter is inserted, not derived. There is no action from which ε_σ emerges as a field obeying equations of motion, so nothing in the present paper distinguishes a uniform ε from a wildly oscillating one beyond aesthetic preference, nothing fixes domain-wall thickness or tension, and nothing addresses stability, whether a given ε -configuration persists, relaxes, or is driven further from uniformity by its own dynamics. Item 4 of the computational programme, a doubled Regge action varied with respect to ε_σ alongside the edge lengths, is the obvious next step and the correct one, but it has not been taken. Until it is, ε is exactly what

it looks like: a hand-set parameter dressed in the vocabulary of a field, and every claim in §5 about where torsion concentrates is conditional on a ε -landscape that is stipulated rather than solved for.

7.2 No physical derivation of the dispersion relation

Equation (4) is motivated geometrically and only geometrically: it is what equation (3) implies once the energy plane is fed through G_d . That motivation establishes internal consistency, that the cross-term is the polarisation of the Gram matrix's off-diagonal entry, but it does not answer the question a particle physicist asks first, which is why the energy-momentum components of a physical state should be regarded as coordinates on a plane measured by the Eisenstein form at all. A dispersion relation is normally derived from a Lagrangian, an equation of motion, or a representation of the Poincaré algebra, none of which are supplied here. The construction shows that *if* the dark sector's kinematic plane carries the Eisenstein Gram matrix, *then* the cross-term follows by polarisation; it does not show why the dark sector's kinematic plane should carry that Gram matrix in the first place. The honest status of equation (4) is therefore a consistent geometric ansatz awaiting a Lagrangian, not a derived physical law, and the paper should not be read as having closed that gap.

What would close these gaps

A field-theoretic completion needs, at minimum: a Lagrangian on the dark sector whose canonical energy-momentum tensor, evaluated on-shell, reproduces equation (4) as its dispersion relation, rather than positing the relation directly; and an action for ε_σ , most naturally the doubled Regge action of programme item 4, whose variation supplies both an equation of motion for the deformation field and, through it, a physical reason for domain-wall formation and stability rather than an assumed ε -landscape. Absent both, this paper's contribution is the geometric skeleton, a norm, a lattice, and the defects they support, not yet the dynamics that would make it a candidate physical theory.

A For the specialist: ε is not a Brans–Dicke scalar

Once ε is described as a field awaiting an action, the natural comparison is to scalar-tensor gravity, and it is worth blocking that comparison explicitly, since it points in the wrong direction. In Brans–Dicke theory and its descendants, a scalar ϕ enters the action through a nonminimal coupling to curvature, a term ϕR , so that ϕ plays the role of a spacetime-dependent gravitational coupling; the theory remains metric-compatible throughout, nonmetricity is zero, and the well-known transformation to the Einstein frame, $g_{\mu\nu}^* = \phi g_{\mu\nu}$, is an isotropic trace-channel rescaling that leaves the physical content unchanged and introduces the scalar's kinetic term only as bookkeeping for the frame change.

The deformation parameter ε does not occupy this role. It does not multiply a curvature scalar, and it is not extracted by any conformal transformation, because, as the shear-versus-trace decomposition of $G(\varepsilon) = I + \varepsilon S$ shows, with S automatically traceless, ε lives entirely in the traceless channel of the Gram matrix, while a Brans–Dicke-type scalar by construction occupies only the trace channel. The two are complementary rather than competing descriptions of the same freedom: nothing prevents a genuine dark-sector Brans–Dicke field $\phi(x)$ from being introduced alongside ε , coupling to whatever plays the role of curvature in the doubled Regge action, with the two fields sourcing orthogonal parts of the dark metric's departure from the baryonic one. Until such a field is introduced, however, this paper contains no gravitational-coupling scalar and no claim about a varying effective Newton constant; ε is closer in spirit to a modulus in the sense of a nonlinear sigma model, a coordinate on the space

of possible norms, than to a Brans–Dicke dilaton, and its prospective action, a double-well potential plus a nearest-neighbour gradient term of the kind sketched in the programme discussion, is accordingly a Landau-type free energy for a shape parameter rather than a kinetic term for a gravitational coupling. The nonmetricity this framework produces is correspondingly of shear type, sourced by $\partial\epsilon$ against the fixed direction S , not the Weyl-vector type a scalar-tensor nonmetricity theory would produce from $\partial \ln \phi$.

B Historical note: the Gram matrix and the metric tensor

The Gram matrix used throughout this paper to encode the dark sector's deformation, and the metric tensor of Riemannian geometry, are the same mathematical object reached by two independent nineteenth-century routes, and the coincidence is worth making explicit since the paper leans on both vocabularies interchangeably. Gauss's first fundamental form [25], the coefficients E, F, G expressing a surface's squared line element as $ds^2 = E du^2 + 2F du dv + G dv^2$, is a symmetric matrix of inner products of the coordinate tangent vectors; Riemann's habilitation lecture [26] generalised this to n dimensions as $ds^2 = \sum g_{ij} dx^i dx^j$, letting the matrix vary smoothly point to point, which is the metric tensor. Independently, and for entirely different reasons, the actuary and mathematician Jørgen Pedersen Gram, working on least-squares approximation and linear independence in abstract inner-product spaces, introduced the determinant and matrix that bear his name [27], with entries $G_{ij} = \langle v_i, v_j \rangle$ for a set of basis vectors. The differential-geometry tradition and the linear-algebra tradition were answering the same question, how does a vector space measure itself internally, from opposite ends: Riemann posits a bilinear form at each point and asks what lengths and angles it implies, while Gram starts from vectors and reads off the bilinear form they carry. The two names persist by convention rather than by any mathematical distinction, "metric" when the matrix is treated as a field over a manifold, "Gram matrix" when it is treated algebraically as the pairing of a fixed basis, and $G(\epsilon)$ in this paper is used in the latter, algebraic sense throughout, becoming a metric proper only if and when ϵ is promoted to a field over the complex, which is precisely the promotion §7 identifies as undone.

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